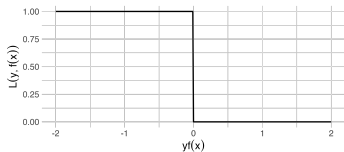
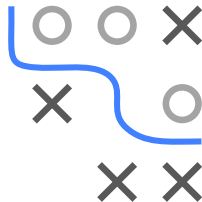


Introduction to Machine Learning

Advanced Risk Minimization Classification and 0-1-Loss



Learning goals

- 0-1 loss
- Risk minimizer(s) / Optim. predictions
- Bayes error rate
- Generative approach in classification

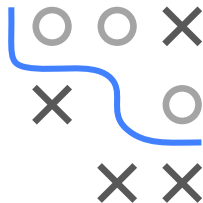
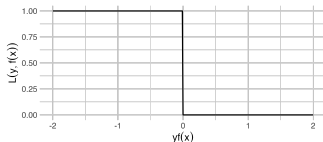
0-1-LOSS

- Discrete classifier

$$h: \mathcal{X} \rightarrow \mathcal{Y}$$

- Maybe most “natural”:
0-1-loss

$$L(y, h(\mathbf{x})) = \mathbb{I}_{\{y \neq h(\mathbf{x})\}}$$



- For $\mathcal{Y} \in \{-1, +1\}$ and scoring classifier $f(\mathbf{x})$
can write it in terms of margin $\nu = yf(\mathbf{x})$

$$L(y, f(\mathbf{x})) = \mathbb{I}_{\{\nu < 0\}} = \mathbb{I}_{\{yf(\mathbf{x}) < 0\}}$$

- For $\mathcal{Y} \in \{0, 1\}$ and prob. classifier $\pi(\mathbf{x})$

$$L(y, \pi(\mathbf{x})) = y\mathbb{I}_{\{\pi(\mathbf{x}) < 0.5\}} + (1-y)\mathbb{I}_{\{\pi(\mathbf{x}) \geq 0.5\}} = \mathbb{I}_{\{(2y-1)(\pi(\mathbf{x})-0.5) < 0\}}$$

- Analytical properties: Not continuous, even for linear $f(\mathbf{x})$ optim.
problem NP-hard = close to intractable ▶ Feldman et al. 2012

RISK MINIMIZER FOR DISCRETE CLASSIFIERS

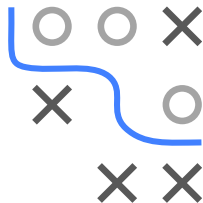
- Again, unravel with law of total expectation (here for multiclass)

$$\begin{aligned}\mathcal{R}(f) &= \mathbb{E}_{\mathbf{xy}} [L(y, f(\mathbf{x}))] = \mathbb{E}_{\mathbf{x}} [\mathbb{E}_{y|\mathbf{x}} [L(y, f(\mathbf{x})) \mid \mathbf{x}]] \\ &= \mathbb{E}_{\mathbf{x}} \left[\sum_{k \in \mathcal{Y}} L(k, f(\mathbf{x})) \mathbb{P}(y = k \mid \mathbf{x}) \right]\end{aligned}$$

- $\eta_k(x) := \mathbb{P}(y = k \mid \mathbf{x})$ is true posterior probability for class k
- For binary case, we denote $\eta(\mathbf{x}) := \mathbb{P}(y = 1 \mid \mathbf{x})$ and get:

$$\mathcal{R}(f) = \mathbb{E}_{\mathbf{x}} [L(1, \pi(\mathbf{x})) \cdot \eta(\mathbf{x}) + L(0, \pi(\mathbf{x})) \cdot (1 - \eta(\mathbf{x}))]$$

- Above formulas work for any loss, not only 0-1; and hard labelers, scorers or prob. classifiers
- Especially for hard labelers and arbitrary misclassif. costs, we see: produces cost-optimal decision, weighted by posterior probs; (we see this again in cost-senslearning)



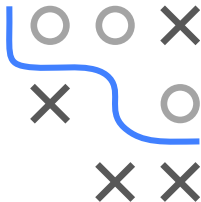
0-1-LOSS: OPTIMAL PREDICTIONS

- For multiclass and hard labeler $h(\mathbf{x})$
- Optimal constant

$$\begin{aligned} h_c^* &= \arg \min_{k \in \mathcal{Y}} \sum_{l \in \mathcal{Y}} L(l, k) \cdot \mathbb{P}(y = l) \\ &= \arg \min_{k \in \mathcal{Y}} \sum_{k \neq l} \mathbb{P}(y = l) \\ &= \arg \min_{k \in \mathcal{Y}} 1 - \mathbb{P}(y = k) \\ &= \arg \max_{k \in \mathcal{Y}} \mathbb{P}(y = k) \end{aligned}$$

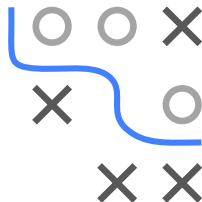
- Translation: Predict most probable class
- Empirical version: $\hat{h}_c = \arg \max_{k \in \mathcal{Y}} \hat{\pi}_k$
- Risk minimizer / optim. cond. prediction / Bayes optim. classifier:

$$h^*(\tilde{\mathbf{x}}) = \arg \max_{k \in \mathcal{Y}} \mathbb{P}(y = k \mid \mathbf{x} = \tilde{\mathbf{x}})$$



BAYES RISK / BAYES ERROR RATE

$$\mathcal{R}^* = 1 - \mathbb{E}_{\mathbf{x}} \left[\max_{k \in \mathcal{Y}} \mathbb{P}(y = k \mid \mathbf{x}) \right]$$



For binary case, can write risk minimizer and Bayes risk as:

$$h^*(\mathbf{x}) = \begin{cases} 1 & \eta(\mathbf{x}) \geq \frac{1}{2} \\ 0 & \eta(\mathbf{x}) < \frac{1}{2} \end{cases}$$

$$\mathcal{R}^* = \mathbb{E}_{\mathbf{x}} [\min(\eta(\mathbf{x}), 1 - \eta(\mathbf{x}))] = 1 - \mathbb{E}_{\mathbf{x}} [\max(\eta(\mathbf{x}), 1 - \eta(\mathbf{x}))]$$

GENERATIVE CLASSIFIERS

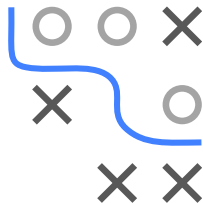
- So, $\arg \max_{k \in \mathcal{Y}} \mathbb{P}(y = k \mid \mathbf{x} = \tilde{\mathbf{x}})$ is what we want to do
- Assume we can model densities given classes and use Bayes:

$$\mathbb{P}(y = k \mid \mathbf{x} = \tilde{\mathbf{x}}) = \frac{p(\tilde{\mathbf{x}} \mid y = k) \mathbb{P}(y = k)}{p(\tilde{\mathbf{x}})}$$

- Then

$$\arg \max_{k \in \mathcal{Y}} \mathbb{P}(y = k \mid \mathbf{x} = \tilde{\mathbf{x}}) = \arg \max_{k \in \mathcal{Y}} p(\tilde{\mathbf{x}} \mid y = k) \mathbb{P}(y = k)$$

- Then we can estimate these conditional densities and the prior probs, and classify via them
- This idea we will see in so-called “generative approaches” for classification, so in LDA, QDA, etc.



EXAMPLE

- Assume $\mathbb{P}(y = 1) = \frac{1}{2}$
- And conditional densities of x per class as normal

$$\begin{cases} \phi_{\mu_1, \sigma^2}(x) & \text{for } y = 0 \\ \phi_{\mu_2, \sigma^2}(x) & \text{for } y = 1 \end{cases}$$
- Bayes optimal classifier = orange; Bayes error = red

